CS F364: Design & Analysis of Algorithms

Quiz 3 – June 5, 2026

Coverage: Lectures 1–12

> Note
>
> **Instructions:**
>
> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question

Consider the following problem: we are given a sequence of n matrices $A_1, A_2, \dots, A_n$ where A_i has dimensions $d_{i-1} \times d_i$, and we want to compute the product $A_1 \cdot A_2 \cdots A_n$ in the order that minimizes the number of scalar multiplications.

(a) [2 Marks] For the instance $n = 4$ with dimensions $d = [5, 10, 3, 8, 6]$, fill in the DP table $m[i, j]$ (minimum cost) and $s[i, j]$ (optimal split point) for multiplying A_i through A_j.

Fill your answers in the tables below:

i	j	$m[i,j]$	$s[i,j]$
1	1	0	-
2	2	0	-
3	3	0	-
4	4	0	-
1	2		
2	3		
3	4		
1	3		
2	4		
1	4		

(b) [1 Mark] What is the minimum number of scalar multiplications to compute $A_1A_2A_3A_4$?

(c) [1 Mark] Write the optimal fully parenthesized expression.

(d) [1 Mark] State the time and space complexity of the general MCM algorithm for n matrices (in Θ notation).

Solution

Part (a) — DP Table

Recurrence:

$$m[i,j] = \begin{cases} 0 & i = j \\ \min\limits_{i \le k < j} \left(m[i,k] + m[k+1,j] + d_{i-1} \cdot d_k \cdot d_j \right) & i < j \end{cases}$$

Dimensions: $d_0 = 5$, $d_1 = 10$, $d_2 = 3$, $d_3 = 8$, $d_4 = 6$

Length 2:

- $m[1,2] = 5 \cdot 10 \cdot 3 = 150$, $s[1,2] = 1$
- $m[2,3] = 10 \cdot 3 \cdot 8 = 240$, $s[2,3] = 2$
- $m[3,4] = 3 \cdot 8 \cdot 6 = 144$, $s[3,4] = 3$

Length 3:

$$m[1,3] = \min \begin{cases} k=1: & m[1,1] + m[2,3] + 5 \cdot 10 \cdot 8 = 0 + 240 + 400 = 640 \\ k=2: & m[1,2] + m[3,3] + 5 \cdot 3 \cdot 8 = 150 + 0 + 120 = 270 \leftarrow \min \end{cases}$$

$s[1,3] = 2$

$$m[2,4] = \min \begin{cases} k=2: & m[2,2] + m[3,4] + 10 \cdot 3 \cdot 6 = 0 + 144 + 180 = 324 \leftarrow \min \\ k=3: & m[2,3] + m[4,4] + 10 \cdot 8 \cdot 6 = 240 + 0 + 480 = 720 \end{cases}$$

$s[2,4] = 2$

Length 4:

$$m[1,4] = \min \begin{cases} k=1: & m[1,1] + m[2,4] + 5 \cdot 10 \cdot 6 = 0 + 324 + 300 = 624 \\ k=2: & m[1,2] + m[3,4] + 5 \cdot 3 \cdot 6 = 150 + 144 + 90 = 384 \leftarrow \min \\ k=3: & m[1,3] + m[4,4] + 5 \cdot 8 \cdot 6 = 270 + 0 + 240 = 510 \end{cases}$$

$s[1,4] = 2$

Completed table:

i	j	$m[i,j]$	$s[i,j]$
1	1	0	-
2	2	0	-
3	3	0	-
4	4	0	-
1	2	150	1
2	3	240	2
3	4	144	3
1	3	270	2
2	4	324	2
1	4	**384**	**2**

Part (b) — Minimum Cost

$$\boxed{\text{384 scalar multiplications}}$$

Part (c) — Optimal Parenthesization

Using $s[1,4] = 2$: split at $k = 2 \to (A_1A_2)(A_3A_4)$

$$\boxed{(A_1A_2)(A_3A_4)}$$

Verification: A_1A_2 costs $5 \cdot 10 \cdot 3 = 150$ (result 5×3), A_3A_4 costs $3 \cdot 8 \cdot 6 = 144$ (result 3×6), final multiply $5 \cdot 3 \cdot 6 = 90$. Total $150 + 144 + 90 = 384$

Part (d) — Complexity

- **Time:** $\Theta(n^3)$ — $O(n^2)$ subproblems, each takes $O(n)$ to evaluate all k.
- **Space:** $\Theta(n^2)$ for the m and s tables.